MATH 223: Linear Algebra

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Abstract

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1 Introduction

2 Prerequisite Knowledge

2.1 Introduction to Proof

2.1.1 Direct Proof

The basic idea of a direct proof of $P \implies Q$ is:

1. Assume P is true,
2. Use P to show that Q is true (the hard part).

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a, b are consecutive integers, meaning $b = a + 1$. So

$$a + b = a + (a + 1) = 2a + 1.$$

Thus b is odd if and only if a is odd. □

2.1.2 Proof by Contradiction

The idea is that a proposition is either true or false, but not both. We get a contradiction when we can show a statement is both true and false, showing our initial assumptions are inconsistent. We use this to show $P \implies Q$ by assuming both P and not Q are simultaneously true and deriving a contradiction.

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a and b are consecutive integers. Assume that $(a + b)$ is not odd. If $(a + b)$ is not odd, then no integer k such that $(a + b) = 2k + 1$. But $a + b = a + (a + 1) = 2a + 1$. We have shown that $(a + b) \neq 2k + 1$ but $(a + b) = 2a + 1$, a contradiction. □

2.1.3 Proof by Contrapositive

The idea is to prove $P \implies Q$ by instead proving not $Q \implies$ not P.

Example. *If n^2 is even, then n is even.*

Proof. We show the contrapositive: if n is odd, then n^2 is odd. Assume n is odd, $\exists k_1 \in \mathbb{Z}, n = 2k_1 + 1$. So

$$n^2 = (2k_1 + 1)^2 = 4k_1^2 + 4k_1 + 1 = 2(2k_1^2 + 2k_1) + 1.$$

Let $k_2 = 2k_1^2 + 2k_1$. Then $n^2 = 2k_2 + 1$, so n^2 is odd. □

2.2 Notation

2.2.1 Sets

Sets are a grouping of objects.

Set	Meaning	Examples
$\mathbb{N}$	The set of natural numbers	(0, 1, 2, 3, ...)
$\mathbb{Z}$	The set of integers	(..., -3, -2, -1, 0, 1, 2, 3, ...)
$\mathbb{Q}$	The set of rational numbers	$\mathbb{Q} = \frac{a}{b} \mid \forall a, b \in \mathbb{Z}\ and\ b \neq 0$
$\mathbb{R}$	The set of all rational and all irrational numbers	$(..., -1, 0, \frac{1}{4}, 1, 1000, ...)$
$\mathbb{C}$	The set of all complex numbers	$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\ and\ i \subseteq \sqrt{-1}\}$.

We have the following relationships between sets:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

2.2.2 Symbols

Symbol	Meaning
$\subseteq$	is a subset of or equal to
$\subset$	is a strict subset of
$\in$	is an element of
$\forall$	for all
$\exists$	there exists
$\emptyset$	empty set
$\Rightarrow$	implies
$\Leftrightarrow$	if and only if
$\cong$	is isomorphic to[1]

2.3 Complex Numbers

2.3.1 Definition and the Fundamental Theorem of Algebra

Definition 1 (Complex Number)**.** *A complex number is of the form: $z = x + iy$ where $x, y \in \mathbb{R}$ and i is the imaginary unit $i = \sqrt{-1}$.*

Theorem 1 (Fundamental Theorem of Algebra)**.** *The FTA states that any non-constant, single-variable polynomial[2] with complex coefficients has at least one root in $\mathbb{C}$.*

Remark 1. *If we have a polynomial f of degree n, then it has n roots, where each root can have a multiplicity[3].*

[1] is isomorphic to = structurally the same as.

[2] A polynomial is a function such as $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ where $a_i \in \mathbb{R}$ or $\mathbb{C}$ and $n \in \mathbb{N}$.

[3] The multiplicity of a root represents how many times the root occurs in the polynomial.

Example. $(x-1)^2$ *has degree 2 but only one root, which is 1, with multiplicity 2.*

We can factorize a polynomial $f = a_n x^n + \cdots + a_1 x + a_0$ as $f = a(x-z_1)(x-z_2)\cdots(x-z_n)$, where z_i are the roots of f in $\mathbb{C}$.

2.3.2 Arithmetic Operations

We define the following operations on complex numbers $z = x + iy$ and $z' = x' + iy'$:

- Addition: $z + z' = (x + x') + i(y + y')$.
- Multiplication: $zz' = (xx' - yy') + i(xy' + yx')$.
- Inverse: $\dfrac{1}{z} = \dfrac{\overline{z}}{z\overline{z}} = \dfrac{x - iy}{x^2 + y^2} = \dfrac{x}{x^2 + y^2} + i\dfrac{-y}{x^2 + y^2}$.

Multiplying by a complex number z corresponds geometrically to:

$$\begin{cases} \text{a rotation by angle } \theta, \\ \text{a rescaling by factor } |z|. \end{cases}$$

2.3.3 Complex Conjugate

Definition 2 (Complex Conjugate)**.** *For $z = x + iy$, the complex conjugate is $\overline{z} = x - iy$.*

Basic properties:

- $\overline{\overline{z}} = z$
- $\overline{z + z'} = \overline{z} + \overline{z'}$
- $\overline{z \cdot z'} = \overline{z} \cdot \overline{z'}$

2.3.4 Geometric Interpretation and Modulus

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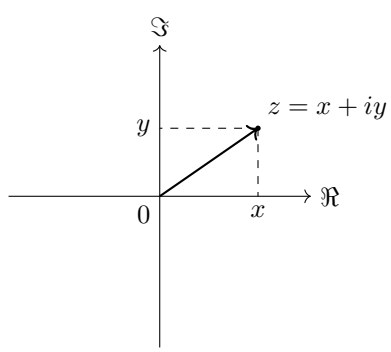
Definition 3 (Geometric interpretation)**.** *Every complex number $z = x + iy$ is identified with a point (x, y) in the complex plane:*

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}.$$



Definition 4 (Modulus)**.** *The modulus of z is*

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

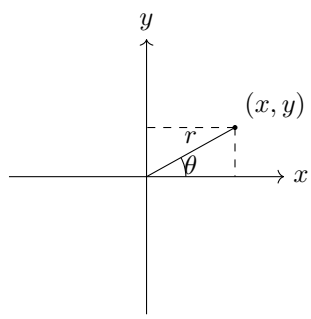
Geometrically, $|z|$ is the distance from the origin to (x, y).

The unit circle in the complex plane can be written as:

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} = \{z \in \mathbb{C} : |z| = 1\}.$$

2.3.5 Polar Coordinates and Polar Form

Definition 5 (Polar coordinates)**.** *Instead of (x, y), a point can be described by polar coordinates (r, θ), where r is the distance to the origin and θ is the angle with the positive x-axis.*



So $x = r\cos\theta$ and $y = r\sin\theta$, with $r = |z|$.

Definition 6 (Euler's formula)**.**

$$e^{i\theta} = \cos\theta + i\sin\theta. \tag{1}$$

Definition 7 (Polar and exponential form)**.** *If $z = x + iy$ with $r = \sqrt{x^2 + y^2}$, then $x = r\cos\theta$, $y = r\sin\theta$, so*

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}.$$

Definition 8 (Multiplication in polar form)**.**

$$z = re^{i\theta}, \quad z' = r'e^{i\theta'} \implies zz' = rr'e^{i(\theta+\theta')}.$$

Example.

$$(1 + i)^{32} = (\sqrt{2}\, e^{i\pi/4})^{32} = 2^{16}\, e^{i8\pi} = 2^{16}(\cos 8\pi + i\sin 8\pi) = 2^{16}.$$

2.3.6 Roots in $\mathbb{C}$

Definition 9 (n^{th} roots)**.** *For $z \in \mathbb{C}$, an nth root of z is $w \in \mathbb{C}$ with $w^n = z$.*

Example. *If $z = re^{i\theta}$, the solutions of $w^n = z$ are*

$$w_k = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, \ldots, n-1.$$

They are equally spaced on a circle of radius $r^{1/n}$ centered at the origin.

Definition 10 (Roots of unity)**.** *The nth roots of unity are solutions of $w^n = 1$:*

$$w_k = e^{i2\pi k/n}, \quad k = 0, 1, \ldots, n-1.$$

They lie on the unit circle and are equally spaced.

2.3.7 Hermiticity

Definition 11 (Hermitian Matrix)**.** *A matrix $A \in M_n(\mathbb{C})$ is Hermitian if $A = A^*$, where $A^* = \overline{A}^T$. Equivalently,*

$$a_{ij} = \overline{a_{ji}} \quad \forall i, j.$$

Example. $A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$ *is Hermitian since $A^* = A$.*

Remark 2. *All eigenvalues of a Hermitian matrix are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Problem 1 (Hermitian Matrix)**.** *Let $A, B \in M_n(\mathbb{C})$. Prove that $(AB)^* = B^* A^*$.*

Definition 12 (Anti-Hermitian Matrix)**.** *A matrix $A \in M_n(\mathbb{C})$ is anti-Hermitian if $A = -A^*$, where $A^* = \overline{A}^T$. Equivalently,*

$$a_{ij} = -\overline{a_{ji}} \quad \forall i, j.$$

Problem 2 (Hermiticity and Anti-Hermiticity)**.** *Prove that A is Hermitian iff iA is anti-Hermitian.*

2.4 Basic Algebraic Structures

2.4.1 Sets with Multiplication

Definition 13 (Set with multiplication)**.** *A set M is a set with multiplication if for any $a, b \in M$, the product $ab \in M$.*

Example. *$M_2(\mathbb{C})$ (all 2×2 complex matrices) and $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ are sets with multiplication.*

2.4.2 Invertibility

Definition 14 (Invertibility)**.** *There are two methods of defining invertibility:*

- *Let $A \in M_n(k)$. If there exists $B \in M_n(k)$ with $AB = I_n$ or $BA = I_n$, then A is invertible and $B = A^{-1}$. Here I_n is the $n \times n$ identity matrix*[4]*.*
- *Let $A \in M_n(k)$ with $\det(A) \neq 0$. Then the inverse of A is*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A),$$

where $\operatorname{adj}(A)$ is the adjugate of A.

Definition 15 (Adjugate of a Matrix)**.** *The adjugate of $A \in M_n(k)$ is the transpose of the cofactor matrix of A:*

$$\operatorname{adj}(A) = \operatorname{cof}(A)^t,$$

where cof *is the cofactor matrix. For a 2×2 matrix this reduces to:*

$$\operatorname{adj}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Definition 16 (Cofactor Matrix)**.** *The cofactor matrix of $A \in M_n(k)$ is the matrix C whose (i, j) entry is the cofactor*

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the determinant of the matrix obtained by deleting row i and column j from A. The sign pattern $(-1)^{i+j}$ is:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Example. *Let* $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$. *The cofactors are:*

$$C_{11} = (+1)\det\begin{pmatrix} 4 & 5 \\ 0 & 6 \end{pmatrix} = 24, \quad C_{12} = (-1)\det\begin{pmatrix} 0 & 5 \\ 1 & 6 \end{pmatrix} = 5,$$

$$C_{13} = (+1)\det\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix} = -4, \quad C_{21} = (-1)\det\begin{pmatrix} 2 & 3 \\ 0 & 6 \end{pmatrix} = -12,$$

$$C_{22} = (+1)\det\begin{pmatrix} 1 & 3 \\ 1 & 6 \end{pmatrix} = 3, \quad C_{23} = (-1)\det\begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} = 2,$$

$$C_{31} = (+1)\det\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = -2, \quad C_{32} = (-1)\det\begin{pmatrix} 1 & 3 \\ 0 & 5 \end{pmatrix} = -5,$$

$$C_{33} = (+1)\det\begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} = 4.$$

[4] The identity matrix has 1s on its diagonal and 0s elsewhere.

The cofactor matrix is

$$\operatorname{cof}(A) = \begin{pmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{pmatrix}$$

Transposing the cofactor matrix to obtain the adjugate:

$$\operatorname{adj}(A) = \begin{pmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{pmatrix}.$$

One can verify: $\det(A) = 22$*, and indeed* $A^{-1} = \frac{1}{22}\operatorname{adj}(A)$.

Remark 3. *If A is invertible, A^{-1} is unique*[5]*.*

Example. $2 \in \mathbb{Z}$ *is* not *invertible since* $\frac{1}{2} \notin \mathbb{Z}$*, but* $2 \in \mathbb{R}$ *is invertible since* $\frac{1}{2} \in \mathbb{R}$.

Example. $1 + i \in \mathbb{C}$ *is invertible:* $\dfrac{1}{1+i} = \dfrac{1-i}{2} \in \mathbb{C}$.

Problem 3 (Invertibility)**.** *Show that if an inverse of A in M exists, then it is unique.*

Problem 4 (Invertibility 2)**.** *Let K be a field. Prove that* $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \in M_2(K)$ *is not invertible.*

2.4.3 Rings

Definition 17 (Ring)**.** *A ring is a set R with addition and multiplication satisfying:*

1. $(R, +)$[6] *is a commutative*[7] *and associative*[8] *group with zero ($\exists 0 \in R$) and additive inverses (every element has an additive inverse $-a$ with $a + (-a) = 0$). Essentially, R behaves like $\mathbb{Z}$ under addition.*

2. *Multiplication is associative with identity* 1*. This means $(ab)c = a(bc)$ for all $a, b, c \in R$, and there exists $1 \in R$ with $1 \cdot a = a \cdot 1 = a$. Note that multiplication need* not *be commutative (i.e., ab may not equal ba), and nonzero elements need not have multiplicative inverses.*

3. *Distributivity: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.*

Example. $(\mathbb{Z}, +, \cdot)$ *is a ring: it is closed under addition and multiplication, has zero element and additive inverses, multiplication is associative with identity 1, and multiplication distributes over addition.*

[5] Unique: there is exactly one such element.

[6] The notation $(R, +)$ means the set R equipped with the operation $+$. It specifies which operation we are talking about, since R has both $+$ and $\times$ defined on it.

[7] $a + b = b + a$.

[8] $a + (b + c) = (a + b) + c$.

2.4.4 Fields

Definition 18 (Field)**.** *A field is a ring* (17) *in which every nonzero element has a multiplicative inverse.*

Example. $(\mathbb{Q}, +, \cdot)$ *is a field: it is a ring, and every nonzero* $q \in \mathbb{Q}$ *has* $q^{-1} \in \mathbb{Q}$ *with* $q \cdot q^{-1} = 1$.

Remark 4. $\mathbb{Z}$ *is a ring but not a field: for example, there is no integer* x *with* $2x = 1$.

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3 Vector Spaces

Definition 19 (Vector Space)**.** *Let* k *be a field. A vector space over* k *is a set* V *with addition and scalar multiplication satisfying:*

1. *Closure under addition:* $v_1 + v_2 \in V$
2. *Commutativity:* $v_1 + v_2 = v_2 + v_1$
3. *Associativity:* $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$
4. *Zero vector:* $\exists\, 0 \in V$ *with* $v + 0 = v$
5. *Additive inverses:* $\forall v \in V, \exists\, -v \in V$ *with* $v + (-v) = 0$
6. *Closure under scalar multiplication:* $\lambda v \in V$
7. *Multiplication associativity:* $\lambda(\mu v) = (\lambda\mu)v$
8. *Identity:* $1 \cdot v = v$
9. *Distributivity over vectors:* $\lambda(v_1 + v_2) = \lambda v_1 + \lambda v_2$
10. *Distributivity over scalars:* $(\lambda + \mu)v = \lambda v + \mu v$

Remark 5. *The axioms for a ring and a vector space look similar because they share the same additive structure. The key difference is that a ring has an internal multiplication (elements of* R *multiply each other), whereas a vector space has an external scalar multiplication (elements of* V *are scaled by elements of a separate field* k*). A field is then just a ring where every nonzero element also has a multiplicative inverse, which is exactly what makes it a suitable set of scalars for a vector space.*

Remark 6. *We often say* abstract *vector space to emphasize that elements of* V *can be any objects (functions, tensors, matrices, etc.) satisfying these axioms — not just arrows or tuples in* $\mathbb{R}^n$.

Example (Non-standard vector space)**.** *Let $V = \{x \in \mathbb{R} : x > 0\}$ be the set of positive reals, with operations:*

$$u \oplus v := uv \quad (\text{“addition” is multiplication}), \qquad \lambda \odot u := u^\lambda \quad (\text{“scalar mult” is exponentiation}).$$

This looks nothing like a normal vector space, so we need to identify what the zero vector and additive inverses are before checking the axioms.
Zero vector. *We need $u \oplus 0_V = u$, i.e., $u \cdot 0_V = u$, so $0_V = 1$.*
Additive inverse. *We need $u \oplus (-u) = 0_V = 1$, i.e., $u \cdot (-u) = 1$, so $(-u) = 1/u$.*
Key axioms. *The non-obvious ones to check are:*

- *Distributivity over scalars: $(\lambda+\mu) \odot u = u^{\lambda+\mu} = u^\lambda \cdot u^\mu = (\lambda \odot u) \oplus (\mu \odot u)$.* ✓
- *Distributivity over vectors: $\lambda \odot (u \oplus v) = (uv)^\lambda = u^\lambda v^\lambda = (\lambda \odot u) \oplus (\lambda \odot v)$.* ✓
- *Scalar compatibility: $\lambda \odot (\mu \odot u) = \lambda \odot (u^\mu) = (u^\mu)^\lambda = u^{\lambda\mu} = (\lambda\mu) \odot u$.* ✓

All 10 axioms hold, so V is a vector space. The reason this works is that V is really just $\mathbb{R}$ in disguise: the map $\varphi : \mathbb{R} \to V$, $x \mapsto e^x$ converts ordinary addition into multiplication and ordinary scalar multiplication into exponentiation.

3.1 Vectors and Linear Combinations

Vector operations in $\mathbb{R}^n$ are defined component-wise:

- Addition: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$.
- Scalar multiplication: $\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$, where $\lambda \in \mathbb{R}$.

Definition 20 (Linear combination)**.** *A linear combination of $v_1, \ldots, v_n$ is $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ for scalars $\lambda_i \in k$.*

Example (Computing a linear combination)**.** *Let $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$.*

Express $v = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$ as a linear combination of v_1 and v_2.
We need $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $\lambda_1 v_1 + \lambda_2 v_2 = v$, i.e.,

$$\lambda_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} \implies \begin{cases} \lambda_1 + 3\lambda_2 = 7 \\ 2\lambda_1 - \lambda_2 = 3. \end{cases}$$

From the second equation $\lambda_2 = 2\lambda_1 - 3$. Substituting: $\lambda_1 + 3(2\lambda_1 - 3) = 7 \implies 7\lambda_1 = 16 \implies \lambda_1 = \frac{16}{7}$, $\lambda_2 = \frac{11}{7}$. Hence

$$v = \tfrac{16}{7} v_1 + \tfrac{11}{7} v_2.$$

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3.2 Span and Standard Basis

3.2.1 Span

Definition 21 (Span)**.** *Let $A \subset \mathbb{R}^n$. The span of A is the set of all linear combinations of elements of A. If $A = \{v_1, \dots, v_k\}$, then*

$$\operatorname{span}(A) = \{\lambda_1 v_1 + \cdots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

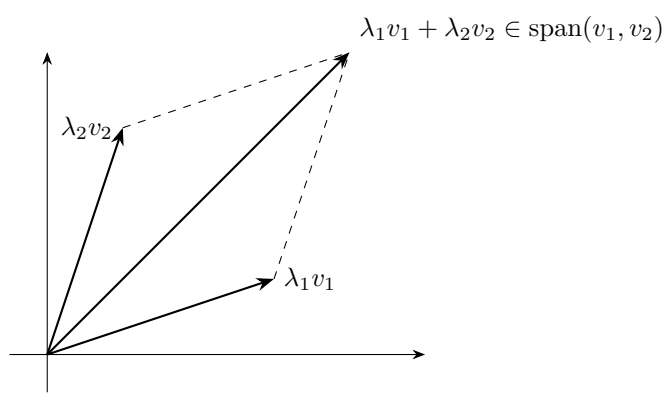
The span describes all points reachable by scaling and adding the given vectors: it can be a line, plane, or higher-dimensional subspace depending on the vectors.

Remark 7. *If $A = \{v\} \subset \mathbb{R}^2$ with $v \neq 0$, then $\operatorname{span}(v) = \{\lambda v : \lambda \in \mathbb{R}\}$ is the line through the origin in the direction of v.*

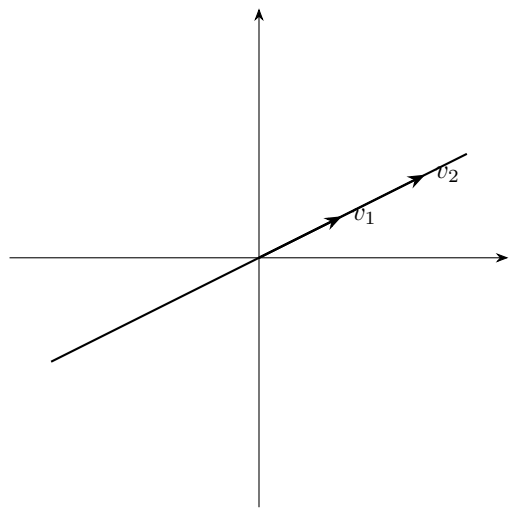
Example. $\operatorname{span}\begin{pmatrix}1\\1\end{pmatrix} = \left\{ t\begin{pmatrix}1\\1\end{pmatrix} \mid t \in \mathbb{R} \right\}$ *is a line in $\mathbb{R}^2$.*

Problem 5 (Span)**.** *Let $v_1 = \begin{pmatrix}3\\1\end{pmatrix}$, $v_2 = \begin{pmatrix}1\\3\end{pmatrix}$. Find $\operatorname{span}(v_1, v_2)$.*

For two linearly independent vectors $v_1, v_2 \in \mathbb{R}^2$, any vector in $\operatorname{span}(v_1, v_2)$ has the form $\lambda_1 v_1 + \lambda_2 v_2$:



If v_1 and v_2 are linearly dependent ($v_1 = \lambda v_2$), then $\operatorname{span}(v_1, v_2)$ is just a line:



Problem 6 (Span 2)**.** *Determine whether the first vector is in the span of the others. If so, write it as a linear combination.*

$$u = (2, 10, 7, 0), \quad u_1 = (3, 10, 7, 0), \quad u_2 = (1, 3, -2, 0), \quad u_3 = (2, 8, 1, 0); \quad u, u_i \in \mathbb{R}^4.$$

3.2.2 Span in $\mathbb{C}^n$

Definition 22 (Span over $\mathbb{C}$)**.**

$$\operatorname{span}_{\mathbb{C}}(v_1, \ldots, v_n) = \{\lambda_1 v_1 + \cdots + \lambda_n v_n : \lambda_i \in \mathbb{C}\}.$$

Example. *Let* $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ i \end{pmatrix} \in \mathbb{C}^2$. *Then* $\lambda_1 v_1 + \lambda_2 v_2 = \begin{pmatrix} \lambda_1 \\ i\lambda_2 \end{pmatrix}$. *Since* λ_1 *and* $i\lambda_2$ *can be any complex number,* $\operatorname{span}_{\mathbb{C}}(v_1, v_2) = \mathbb{C}^2$.

3.2.3 Standard Basis

Definition 23 (Standard basis of $\mathbb{R}^n$)**.** *The standard basis is* $\{e_1, \ldots, e_n\}$, *where* e_i *has a 1 in the* i*-th coordinate and 0 elsewhere:*

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Equivalently, $e_1, \ldots, e_n$ *are the columns of* I_n.

Proposition 1. *Every* $x \in \mathbb{R}^n$ *can be written uniquely as* $x = x_1 e_1 + \cdots + x_n e_n$.[9]

[9] To verify the standard basis is indeed a basis, see Appendix (9.1).

The standard basis also clarifies the difference between real and complex vector spaces — the same vectors generate very different spans depending on the scalar field.

Example (Real vs complex span)**.** *Using e_1, e_2 in $\mathbb{C}^2$: over field $\mathbb{C}$, $\operatorname{span}_{\mathbb{C}}(e_1, e_2) = \mathbb{C}^2$ with $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$. Over field $\mathbb{R}$, e_1, e_2 cannot produce vectors with imaginary parts. To span $\mathbb{C}^2$ over $\mathbb{R}$, we also need ie_1, ie_2, so $\operatorname{span}_{\mathbb{R}}(e_1, e_2, ie_1, ie_2)$, giving*

$$\dim_{\mathbb{R}} \mathbb{C}^2 = 4.$$

3.3 Examples of Vector Spaces

14 January, 2026

3.3.1 Coordinate Spaces

Definition 24 (Coordinate space)**.**

$$k^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in k \right\}, \quad \dim k^n = n.$$

Standard basis: $\{e_1, \ldots, e_n\}$.

3.3.2 Polynomial Spaces

Definition 25 (Polynomial space)**.**

$$P_n(k) = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in k\}, \quad \dim P_n(k) = n + 1.$$

Standard basis: $\{1, x, x^2, \ldots, x^n\}$, with $P_n(k) \cong k^{n+1}$. The isomorphism is simply the map that reads off the coefficients:

$$a_0 + a_1 x + \cdots + a_n x^n \longleftrightarrow \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix}.$$

A polynomial is completely determined by its $n + 1$ coefficients, so it carries exactly the same information as a vector in k^{n+1}.

Example. *$1 + x^2 \in P_2(\mathbb{R})$, which has standard basis $\{1, x, x^2\}$ and $\dim P_2(\mathbb{R}) = 3$.*
$1 + ix^3 \in P_3(\mathbb{C})$, which has standard basis $\{1, x, x^2, x^3\}$ and $\dim P_3(\mathbb{C}) = 4$.

Remark 8. *The space of all polynomials $P_\infty(k) = \{a_0 + a_1 x + \cdots : a_i \in k\}$ is infinite-dimensional, and the spaces form a nested sequence*

$$P_0(k) \subseteq P_1(k) \subseteq \cdots \subseteq P_\infty(k).$$

3.3.3 Matrix Spaces

Definition 26 (Matrix space)**.**

$$M_n(k) = \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \mid a_{ij} \in k \right\}, \quad \dim M_n(k) = n^2.$$

Standard basis: $\{e_{ij} : 1 \leq i, j \leq n\}$, where e_{ij} has a 1 in the (i, j)-th entry and 0 elsewhere.

Example. *In $M_2(\mathbb{R})$: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1e_{11} + 2e_{12} + 3e_{21} + 4e_{22}$.*

3.3.4 Function Spaces

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Definition 27 (Function space)**.** *Let D be a set and k a field. Define*

$$F(D, k) = \{f : D \to k\},$$

with pointwise operations $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$. This is a vector space over k.

Finite case. If $D = \{1, \ldots, n\}$, define the Kronecker delta functions

$$\delta_i(j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $\{\delta_1, \ldots, \delta_n\}$ is a basis of $F(D, k)$ and every f can be written uniquely as $f = \sum_{i=1}^{n} f(i)\, \delta_i$. Hence $\dim F(D, k) = n$.

Infinite case. If D is infinite, $F(D, k)$ is infinite-dimensional: there is no finite spanning set. The natural candidate for a basis, $\{\delta_x : x \in D\}$, is linearly independent but fails to span because linear combinations must be finite — a general $f : D \to k$ can be nonzero at infinitely many points and cannot be expressed as a finite sum of δ_x's.

Example. *Let $D = \{1, 2, 3\}$, $k = \mathbb{R}$, and f with $f(1) = 2$, $f(2) = -1$, $f(3) = 3$. The standard basis is $\{\delta_1, \delta_2, \delta_3\}$ and $f = 2\delta_1 - \delta_2 + 3\delta_3$.*

3.4 Subspaces

January 19, 2026

3.4.1 Definition and the Subspace Criterion

Definition 28 (Subspace)**.** *A subspace of V is a subset $U \subseteq V$ that is itself a vector space over the same field k.*

Proposition 2 (Subspace criterion)**.** *$U \subseteq V$ is a subspace if and only if:*

1. $0 \in U$,
2. $u, v \in U \Rightarrow u + v \in U$,
3. $u \in U$, $\lambda \in k \Rightarrow \lambda u \in U$.

Remark 9 (Proving a vector space in practice)**.** *To prove that a set U is a vector space over k, you do* ***not*** *always need to verify all 10 axioms. If U is a subset of an already-known vector space V over the same field k, then 7 of the 10 axioms are* inherited automatically *from V. In that case, it suffices to verify only the 3 subspace criteria. The full 10-axiom verification is only necessary when constructing a completely new vector space with no ambient space to inherit from.*

Example (Proving a subspace with the 3-criteria)**.** *Let $V = \mathbb{R}^3$ and $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Prove U is a subspace.*
Since $U \subset \mathbb{R}^3$, we only check the 3 criteria.

1. ***Zero****: $0 + 0 + 0 = 0$, so $(0, 0, 0) \in U$.* ✓
2. ***Addition****: Let $u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2) \in U$. Then*
$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0.$$
so $u + v \in U$. ✓
3. ***Scalar mult****: $(\lambda x) + (\lambda y) + (\lambda z) = \lambda(x + y + z) = 0$, so $\lambda u \in U$.* ✓

Hence U is a subspace of $\mathbb{R}^3$. Note that U is a plane through the origin.

Problem 7 (Subspace criterion)**.** *Let $A \in M_n(k)$. Prove that $U = \{x \in k^n : Ax = \vec{0}\}$ is a subspace (the null space or kernel of A).*

Problem 8 (Subspace criterion 2)**.** *Let $A \in M_n(k)$. Show that $U = \{Ax \mid x \in k^n\}$ is a subspace of k^n (the image of A).*

Problem 9 (Subspace criterion 3)**.** *Let $V = k$, viewed as a vector space over k. Show that the only subspaces are $\{0\}$ and V itself.*

Problem 10 (Subspace criterion 4)**.** *Let $v_1, \ldots, v_n \in V$. Show that* $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace of V.*

3.4.2 Membership in a Span

January 21, 2026

Problem 11 (Span membership)**.** *Let* $v = \begin{pmatrix} 3 \\ 5 \\ -5 \end{pmatrix}$, $v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 2 \\ 5 \\ 4 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}$. *Decide whether $v \in \operatorname{span}(v_1, v_2, v_3)$.*

Problem 12 (Span membership 2)**.** *Let $f = 3x^2 + 5x - 5$, $f_1 = x^2 + 2x + 1$, $f_2 = 2x^2 + 5x + 4$, $f_3 = x^2 + 3x + 6$. Decide whether $f \in \operatorname{span}(f_1, f_2, f_3)$.*

3.4.3 Operations on Subspaces

Let W be a vector space and $U, V \subseteq W$ subspaces.

Definition 29 (Intersection)**.**

$$U \cap V = \{w \in W \mid w \in U \text{ and } w \in V\}.$$

The intersection of subspaces is a subspace; the smallest possible is $\{0\}$*.*

Definition 30 (Union)**.**

$$U \cup V = \{w \in W \mid w \in U \text{ or } w \in V\}.$$

The union of subspaces is not *necessarily a subspace.*

Proposition 3. *$U \cup V$ is a subspace of W if and only if $U \subseteq V$ or $V \subseteq U$.*

Proof. If $U \subseteq V$, then $U \cup V = V$, a subspace. Conversely, if neither contains the other, pick $u \in U \setminus V$ and $v \in V \setminus U$. If $U \cup V$ were a subspace, $u+v \in U \cup V$. If $u + v \in U$, then $v = (u + v) - u \in U$, a contradiction. If $u + v \in V$, then $u = (u + v) - v \in V$, also a contradiction. □

Definition 31 (Sum of subspaces)**.**

$$U + V = \{u + v \mid u \in U,\, v \in V\}.$$

Proposition 4. *If U and V are subspaces of W, then $U + V$ is a subspace of W.*

Proof. Apply the subspace criterion: $0 = 0 + 0 \in U + V$; if $w_1 = u_1 + v_1$ and $w_2 = u_2 + v_2$ then $w_1 + w_2 = (u_1 + u_2) + (v_1 + v_2) \in U + V$; for $\lambda \in K$, $\lambda w = \lambda u + \lambda v \in U + V$. □

Proposition 5. *$U + V = \operatorname{span}(U \cup V)$, i.e., $U + V$ is the smallest subspace containing both U and V.*

Proof. ($\subseteq$): Any $u + v \in U + V$ with $u, v \in U \cup V$ lies in $\operatorname{span}(U \cup V)$.
($\supseteq$): Any $w \in \operatorname{span}(U \cup V)$ can be grouped as a sum of a U-part and a V-part, so $w \in U + V$. □

3.4.4 Direct Sums

January 23, 2026

Definition 32 (Direct sum)**.** *We say U and V are in direct sum if $U \cap V = \{0\}$, and write $U \oplus V$ for their sum. We write*

$$W = U \oplus V \iff \begin{cases} W = U + V, \\ U \cap V = \{0\}. \end{cases}$$

Proof. $W = U \oplus V$ iff every $w \in W$ decomposes *uniquely* as $w = u + v$. Uniqueness follows since $u_1 - u_2 = v_2 - v_1 \in U \cap V = \{0\}$ forces $u_1 = u_2$. Conversely, if decompositions are unique and $w \in U \cap V$, then $0 = w + (-w) = 0 + 0$ forces $w = 0$. □

Analogy with sets.

Sets	Vector spaces
$A \cap B$	$U \cap V$
$A \cup B$	$U + V$
$A \sqcup B,\ A \cap B = \emptyset$	$U \oplus V,\ U \cap V = \{0\}$

Direct sums in function spaces.

Definition 33 (Support on a subset)**.** *Let $A \subseteq D$. Define $U = \{f \in F(D, \mathbb{R}) : f(x) = 0\ \forall x \notin A\}$. Then $U \cong F(A, \mathbb{R})$ is a subspace of $F(D, \mathbb{R})$.*

Example. *Let $D = \{1, 2, 3, 4, 5\}$, $A = \{2, 4\}$. Then $U = \{f : f(1) = f(3) = f(5) = 0\}$ is a 2-dimensional subspace, with each $f \in U$ determined by $(f(2), f(4))$.*

4 Basis and Dimension

January 26, 2026

4.1 Finite Dimensional Spaces

Let V be a vector space over a field k. Either $V = \{0\}$, or there exists $v_1 \neq 0$ in V. In the latter case, either $V = \operatorname{span}(v_1)$, or there exists $v_2 \notin \operatorname{span}(v_1)$, and so on. The maximum number of linearly independent vectors is the dimension of V.

Definition 34 (Finite dimensional vector space)**.** *V is finite-dimensional if $\exists$ a finite set $v_1, \ldots, v_n \in V$ with $V = \operatorname{span}(v_1, \ldots, v_n)$. Its dimension is*

$$\dim_k V = \min\{n : V = \operatorname{span}_k(v_1, \ldots, v_n)\}.$$

Example ($\mathbb{C}$ as a real vector space)**.** $\dim_{\mathbb{R}} \mathbb{C} = 2$ *(basis $\{1, i\}$). For $M_n(\mathbb{C})$: $\dim_{\mathbb{C}} M_n(\mathbb{C}) = n^2$ and $\dim_{\mathbb{R}} M_n(\mathbb{C}) = 2n^2$.*

Example (P_∞ is infinite-dimensional)**.** *Since $x^{n+1} \notin P_n = \operatorname{span}\{1, x, \ldots, x^n\}$ for all n, the space $P_\infty(\mathbb{R})$ requires arbitrarily many basis vectors.*

Problem 13 (Infinite dimension of P_∞)**.** *Prove that P_∞ over any field k has infinite dimension.*

4.2 Linear Independence

Definition 35 (Linear independence)**.** *Vectors $v_1, \ldots, v_n$ are linearly independent (LI) if*

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = 0 \implies \lambda_1 = \cdots = \lambda_n = 0.$$

Definition 36 (Linear dependence)**.** *Linearly dependent (LD) is the negation of LI: there exist scalars, not all zero, with $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$.*

Example. $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ *are LI:* $xe_1 + ye_2 = \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x = y = 0$.

Proposition 6. *$v_1, \dots, v_n$ are LD if and only if one vector lies in the span of the others.*

Proof. ($\Rightarrow$) If $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$ with some $\lambda_k \neq 0$, then $v_k = \frac{-1}{\lambda_k} \sum_{i \neq k} \lambda_i v_i$. ($\Leftarrow$) If $v_k = \sum_{i \neq k} \lambda_i v_i$, rearranging gives a nontrivial relation. □

Remark 10. *If any vector $v_i = 0$, the family is LD.*

Example. *For $v_1 = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} b_1 \\ b_2 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$, if $a_1, b_2, c_3 \neq 0$ then v_1, v_2, v_3 are LI (solved by back-substitution). If $c_3 = 0$, all vectors lie in a 2D subspace and are LD.*

January 28, 2026

Problem 14 (Linear independence)**.** *Let $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$, $v_3 = \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix}$.*

Are these LI?

4.3 Basis

4.3.1 Definition and the Main Basis Theorem

Definition 37 (Basis)**.** *$(v_1, \dots, v_n)$ is a basis of V if it spans V and is linearly independent.*

Remark 11. *If V has a basis of n vectors, then $\dim V = n$.*

January 30, 2026

Example (Standard basis of $\mathbb{R}^3$)**.** *$\{e_1, e_2, e_3\}$ is a basis of $\mathbb{R}^3$: it spans (any $v = xe_1 + ye_2 + ze_3$) and is LI (setting to zero gives $x = y = z = 0$).*

Lemma 2. *For $(v_1, \dots, v_n) \subset \mathbb{R}^k$: if $n < k$ the vectors cannot span $\mathbb{R}^k$; if $n > k$ they cannot be LI.*

Lemma 3. *If $u_1, \dots, u_n$ span V and $v_1, \dots, v_m$ are LI in V, then $n \geq m$.*

Theorem 4 (Main Basis Theorem)**.** *$(v_1, \dots, v_n)$ in V are equivalent (these four properties are all secretly the same thing):*

1. ***Basis**: they span V and are LI.*
2. ***Minimal span**: they span V and no $m < n$ vectors span V.*
3. ***Maximal LI**: they are LI and no $m > n$ vectors are LI.*
4. ***Unique coordinates**: every $v \in V$ writes uniquely as $v = \sum_{i=1}^{n} \lambda_i v_i$.*

Corollary 4.1. *Any two bases of V have the same number of vectors. Hence $\dim V$ is well-defined.*

Proof. If $(e_1, \dots, e_n)$ and $(f_1, \dots, f_m)$ are both bases, the two-way spanning/independence argument gives $n \geq m$ and $m \geq n$. □

February 2, 2026

4.3.2 Finding a Basis

There are several methods depending on how the space is described.

Method 1: Extracting from a spanning set.

Use this when you are given explicit vectors in k^m and want to find which ones form a basis.

1. Form a matrix A whose *columns* are the vectors.
2. Row reduce A to echelon form.
3. The original vectors corresponding to pivot columns form a basis.

Problem 15 (Finding a Basis)**.** *Let V be the subspace of $\mathbb{R}^4$ spanned by $v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 2 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$, $v_4 = \begin{pmatrix} 2 \\ 8 \\ 2 \\ 10 \end{pmatrix}$. Find a subset forming a basis of V.*

Method 2: Testing linear independence.

Use this when the vectors live in an abstract space (matrices, polynomials) that you can flatten into coordinates.

1. Map each vector to its coordinate vector in k^n using a known isomorphism.
2. Set up $a_1 v_1 + \cdots + a_n v_n = 0$ and solve.
3. Drop any vector whose coefficient is forced to be free (it is a linear combination of the others).

Problem 16 (Finding a Basis 2)**.** *Find a basis of the span of $M_1 = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, $M_2 = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$, $M_3 = \begin{pmatrix} 4 & 5 \\ 5 & 7 \end{pmatrix}$ in $M_2(\mathbb{R})$.*

Method 3: Constructing from the general form.

Use this when the subspace is defined by constraints (e.g. symmetry, zero entries). Write out the most general element satisfying the constraints and identify the free parameters — each gives one basis vector.

Problem 17 (Finding a Basis 3)**.** *(1) Show $A \in M_2(\mathbb{C})$ is Hermitian iff $A = \begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}$ for $x, y \in \mathbb{R}$, $z \in \mathbb{C}$.*
(2) Find a basis of the set U of all Hermitian matrices in $M_2(\mathbb{C})$, viewed as a real vector space.

Method 4: Bases of function spaces.

Use this when the space is $F(X)$ for a finite set X. The dimension equals the number of free values, and you verify a candidate basis by checking LI (evaluate at each point) and count.

Problem 18 (Finding a Basis 4)**.** *Let $X = \{-2, -1, 0, 1, 2\}$, $U = \{f \in F(X) : f(-x) = f(x)\ \forall x\}$. With $g_1(x) = 1$, $g_2(x) = \frac{1}{4}x^2(1 - x^2)$, $g_3(x) = \frac{1}{3}x^2(4 - x^2)$, prove $S = (g_1, g_2, g_3)$ is a basis of U.*

4.3.3 Proving a Set Forms a Basis

Method 1 (Dimension test)**.** *$(v_1, \ldots, v_n)$ is a basis of V iff $\dim(\operatorname{span}(v_1, \ldots, v_n)) = n$.*

Proposition 7 (Basis criterion)**.** *If $\dim V = n$, then for $v_1, \ldots, v_n \in V$, the following are equivalent: (1) basis, (2) spans V, (3) LI.*

Proof. A spanning set of $n = \dim V$ vectors is minimal, hence LI. An independent set of n vectors is maximal, hence spans V. □

Principle 1 (Transfer/Exchange principle)**.** *If $u_1, \ldots, u_n$ span V and $v_1, \ldots, v_n$ are LI, then both lists are bases of V.*

Example. *In P_n: $(1, x, \ldots, x^n)$ spans and $((x-a)^0, \ldots, (x-a)^n)$ is LI, so both are bases of P_n.*

February 11, 2026

4.3.4 Basis Extension and Extraction

Theorem 5 (Basis extension)**.** *Let V be finite-dimensional and $v_1, \ldots, v_m$ LI in V. Then they can be extended to a basis of V.*

Proof. Set $U = \operatorname{span}(v_1, \ldots, v_m)$. If $U = V$, done. Otherwise pick $v_{m+1} \in V \setminus U$; this gives a larger LI set. Repeat until the span equals V, which happens after at most $\dim V - m$ steps. □

Theorem 6 (Basis extraction)**.** *If $v_1, \ldots, v_n$ span V, one can extract a basis from this list.*

Problem 19 (Basis Extension)**.** *Let V be a finite-dimensional vector space and $L \subseteq V$ linearly independent. Show there is a basis $B \supseteq L$ of V.*

4.4 Coordinates and Change of Basis

February 4–6, 2026

4.4.1 Coordinate Vectors

Definition 38 (Coordinate vectors)**.** *For basis $(v_1, \ldots, v_n)$ of V, every $v = x_1v_1 + \cdots + x_nv_n$ uniquely, and $[v]_e = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ is the coordinate vector of v.*

Example (Finding a coordinate vector)**.** *Find the coordinate vector of* $v = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}$ *in the basis* $f = (f_1, f_2, f_3)$ *where* $f_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $f_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.
We need x, y, z *such that* $xf_1 + yf_2 + zf_3 = v$. *Writing this as a system:*

$$\begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}.$$

Row reduce the augmented matrix:

$$\begin{pmatrix} 1 & 1 & 0 & 5 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2+R_1} \begin{pmatrix} 1 & 1 & 0 & 5 \\ 0 & 2 & 1 & 8 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2-R_3} \begin{pmatrix} 1 & 1 & 0 & 5 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$$

Back-substitution: $z = 4$, $y = 2$, $x = 3$. *Therefore* $[v]_f = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$.

Problem 20 (Coordinate vectors)**.** *Let* $X = \{-2, -1, 0, 1, 2\}$, $U = \{f \in F(X) : f(-x) = f(x)\}$, *with basis* $S = (g_1, g_2, g_3)$ *as above. Find* $[g]_S$ *for* g *with* $g(\pm 2) = 3$, $g(\pm 1) = g(0) = 2$.

Problem 21 (Coordinate vectors 2)**.** *Let* $D = \{1, 2, 3\}$. *Find the coordinates of* $f(x) = 2x^2 - 3$ *in the standard basis of* $F(D)$.

4.4.2 Direct Sum Decompositions and Dimension Formulas

Theorem 7 (Basis concatenation)**.** *If* $V = U \oplus U'$ *with bases* $(e_1, \ldots, e_n)$ *and* $(f_1, \ldots, f_m)$ *respectively, then* $(e_1, \ldots, e_n, f_1, \ldots, f_m)$ *is a basis of* V *and* $\dim(U \oplus U') = \dim U + \dim U'$.

Proof. LI: $\sum a_i e_i + \sum b_j f_j = 0$ gives $w = \sum a_i e_i = -\sum b_j f_j \in U \cap U' = \{0\}$, forcing all $a_i = b_j = 0$. Span: every $v \in U + U'$ writes $v = u + u'$. □

Theorem 8 (Dimension of sum)**.**

$$\dim(U + U') = \dim U + \dim U' - \dim(U \cap U').$$

Corollary 8.1. *If* $U \subseteq V$, *then* $\dim U \leq \dim V$.

Corollary 8.2. *If* $U \subseteq V$, *then* U *admits a complement:* $\exists U' \subseteq V$ *with* $V = U \oplus U'$ *and* $\dim V = \dim U + \dim U'$.

5 Linear Transformations

5.1 Functions: Basic Definitions

Let U and V be sets. A transformation $T : U \to V$ assigns to each $u \in U$ a unique element $T(u) \in V$. The set U is the domain, V the codomain, and $\mathrm{Im}(T) = \{T(u) : u \in U\} \subseteq V$ is the image.

Definition 39 (Injective, surjective, bijective)**.**

- *T is **surjective** if $\mathrm{Im}(T) = V$, i.e., every element of V is hit by at least one element of U.*
- *T is **injective** if $T(u) = T(u') \implies u = u'$, where u' is any other element of U. In other words, no two distinct inputs give the same output — every element of V is hit by* at most *one element of U.*
- *T is **bijective** if it is both injective and surjective, i.e., every element of V is hit by* exactly *one element of U.*

Example. *$f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$ is bijective: every real number has exactly one real cube root, so distinct inputs give distinct outputs (injective) and every output is achieved (surjective). On the other hand, $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$ is neither: it is not injective since $(-2)^2 = 2^2 = 4$ (two inputs, same output), and not surjective since negative numbers are never achieved.*

Example. *$z \mapsto z^2$ on $\mathbb{C}$: not injective since $(-1)^2 = 1^2 = 1$ (so two distinct complex numbers map to the same value). It is however surjective: every $w \in \mathbb{C}$ has a square root in $\mathbb{C}$ by the Fundamental Theorem of Algebra, so every output is achieved.*

5.1.1 Composition and Inverses

Definition 40 (Composition)**.** *Given $T : U \to V$ and $S : V \to W$, the composition $R = S \circ T : U \to W$ is defined by $R(u) = S(T(u))$.*

Example. *$g = \sin \circ (x^2) : \mathbb{R} \xrightarrow{x^2} \mathbb{R} \xrightarrow{\sin} \mathbb{R}$, so $g(x) = \sin(x^2)$.*

Proposition 8 (Inverses)**.** *$T : U \to V$ is bijective if and only if T has an inverse $S : V \to U$ with $ST = \mathrm{id}_U$[10] and $TS = \mathrm{id}_V$, where $\mathrm{id}_U(u) = u$ and $\mathrm{id}_V(v) = v$ are the identity maps[11].*

5.2 Linear Maps

Definition 41 (Linear transformation)**.** *Let U, V be vector spaces over k. A map $T : U \to V$ is **linear** if:*

[10] remember ST means "apply T first, then S."

[11] The identity map sends every element to itself; it is the function analogue of the identity matrix.

1. *Additive:* $T(u+u') = T(u) + T(u')$,
2. *Scalar:* $T(\lambda u) = \lambda T(u)$ *for all* $\lambda \in k$.

5.2.1 Main Examples of Linear Maps

Matrix transformations

The primary example is $T_A : \mathbb{R}^n \to \mathbb{R}^n$, $T_A(x) = Ax$ for a fixed $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \implies T_A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}.$$

T_A is bijective if and only if A is invertible ($\det(A) \neq 0$): if A^{-1} exists, then $T_{A^{-1}}$ is the inverse map, so every output has exactly one input. If $\det(A) = 0$, then $Ax = 0$ has a nonzero solution, meaning T_A collapses an entire direction to zero — it is neither injective (two inputs map to the same output) nor surjective (some outputs are never reached).

Scaling

Definition 42 (Scaling transformation)**.** *$T : U \to U$ is called scaling if there exists a basis $(u_1, \ldots, u_n)$ and scalars $\lambda_1, \ldots, \lambda_n \in k$ such that $T(u_i) = \lambda_i u_i$ for all i.*

T is bijective if all $\lambda_i \neq 0$: each basis vector is simply stretched or shrunk but never sent to zero, so distinct inputs stay distinct (injective) and every output is reachable by scaling back (surjective). If any $\lambda_i = 0$, then $T(u_i) = 0 = T(0)$, so two distinct vectors map to the same output — T is neither injective nor surjective.

The derivative

$$D : P_n \to P_n, \quad f \mapsto f', \quad D(x^k) = kx^{k-1}.$$

D is linear: $(f+g)' = f' + g'$ and $(\lambda f)' = \lambda f'$. D is not injective because any two polynomials differing by a constant have the same derivative, e.g. $D(1) = D(0) = 0$, so $\ker(D)$ contains all constants. D is not surjective because x^n is never achieved: differentiating any polynomial in P_n drops the degree by 1, so nothing in P_n can differentiate to x^n.

The indefinite integral

$$\int : P_\infty \to P_\infty \quad \text{[linear, injective, not surjective]}.$$

Injective: integration raises the degree, so if $\int f = \int g$ then $f = g$ — no two distinct polynomials integrate to the same thing. Equivalently, $\ker(\int) = \{0\}$.

Not surjective: the constant polynomial $1 \in P_\infty$ is never an output, since $\int f$ always has zero constant term (or more precisely, $\int f$ always has degree at least 1).

Projection

Suppose $V = U \oplus U'$. Every $v \in V$ writes uniquely as $v = u + u'$. Define $P(v) := u$.

Proposition 9. *The projection $P : V \to U$ along U' is linear, surjective, but not injective.*

Proof. Linearity: write $v_1 = u_1 + u_1'$ and $v_2 = u_2 + u_2'$. Then $P(v_1 + v_2) = u_1 + u_2 = P(v_1) + P(v_2)$. Scalar case is analogous. Surjective: any $u \in U$ satisfies $P(u) = u$ (since $u = u + 0$ with $0 \in U'$), so every element of U is hit. Not injective: any nonzero $u' \in U'$ satisfies $P(u') = 0 = P(0)$, so two distinct vectors map to the same output. □

5.2.2 Defining a Linear Map by its Action on a Basis

Proposition 10. *Let $(u_1, \dots, u_n)$ be a basis of U and $v_1, \dots, v_n$ arbitrary vectors in V. There exists a unique linear $T : U \to V$ with $T(u_i) = v_i$.*

Proof. Existence: for any $u = \lambda_1 u_1 + \cdots + \lambda_n u_n$, define $T(u) = \lambda_1 v_1 + \cdots + \lambda_n v_n$. Linearity follows from uniqueness of coordinates.
Uniqueness: any linear T with $T(u_i) = v_i$ is forced on all u by linearity. □

Proposition 11. *If S, T are linear, then $S \circ T$ is linear.*

5.3 Kernel and Image

February 18, 2026

Definition 43 (Kernel and image)**.**

$$\ker(T) = \{u \in U : T(u) = 0\}, \qquad \mathrm{Im}(T) = \{T(u) : u \in U\} \subseteq V.$$

Proposition 12. $\ker(T) \subseteq U$ *and* $\mathrm{Im}(T) \subseteq V$ *are subspaces.*

Proposition 13. *T is injective* $\iff$ $\ker(T) = \{0\}$.

Proof. ($\Leftarrow$) If $\ker(T) = \{0\}$ and $T(u) = T(u')$, then $T(u - u') = 0$, so $u - u' = 0$.
($\Rightarrow$) Contrapositive: if $\exists u \neq 0$ with $T(u) = 0$, then $T(u) = T(0)$ with $u \neq 0$. □

Remark 12. *The kernel is only useful for linear T. For example,* $\ker(\sin) = \{k\pi : k \in \mathbb{Z}\}$ *is not a subspace.*

Proposition 14. *If $(e_1, \dots, e_n)$ is a basis of U, then* $\mathrm{Im}(T) = \mathrm{span}(T(e_1), \dots, T(e_n))$.

Proof. Any $v = T(u) = T(\lambda_1 e_1 + \cdots + \lambda_n e_n) = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n) \in \mathrm{span}(T(e_i))$. □

5.3.1 Rank-Nullity Theorem

Theorem 9 (Rank-Nullity Theorem)**.** *Let $T : U \to V$ be a linear map with U finite-dimensional. Then*

$$\dim \ker(T) + \dim \operatorname{Im}(T) = \dim U.$$

Proof. Let $n = \dim U$, $k = \dim \ker(T)$, and let $(u_1, \dots, u_k)$ be a basis of $\ker(T)$. By basis extension, extend to a basis $(u_1, \dots, u_k, u_{k+1}, \dots, u_n)$ of U. We claim $(T(u_{k+1}), \dots, T(u_n))$ is a basis of $\operatorname{Im}(T)$.

Span: Any $T(u) = T(\sum \lambda_i u_i) = \sum_{i=k+1}^n \lambda_i T(u_i)$ since $T(u_i) = 0$ for $i \le k$.

LI: Suppose $\sum_{i=k+1}^n \lambda_i T(u_i) = 0$. Then $T\big(\sum_{i=k+1}^n \lambda_i u_i\big) = 0$, so $\sum_{i=k+1}^n \lambda_i u_i \in \ker(T) = \operatorname{span}(u_1, \dots, u_k)$. Since $(u_1, \dots, u_n)$ is LI, all $\lambda_i = 0$.

Hence $\dim \operatorname{Im}(T) = n - k$, which gives the result. □

Remark 13. $\dim \operatorname{Im}(T)$ *is called the* ***rank*** *of T and* $\dim \ker(T)$ *the* ***nullity****. For a matrix $A \in M_{m\times n}(k)$, rank-nullity says: rank(A) + nullity(A) = n (number of columns).*

5.3.2 Examples of Kernel and Image

Example. *For the projection $P : V \to U$ along U':* $\operatorname{Im}(P) = U$ *and* $\ker(P) = U'$.

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $T_A : \mathbb{R}^2 \to \mathbb{R}^2$. *Solving* $\begin{cases} x + 2y = 0 \\ 3x + 4y = 0 \end{cases}$ *gives* $x = y = 0$, *so* $\ker(T_A) = \{0\}$ *and T_A is injective.*

Example. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. *Row reduction gives* $\ker(T_A) = \operatorname{span}\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.

February 20, 2026

5.4 Matrix of a Linear Transformation

Definition 44 (Matrix of T in basis e)**.** *Let $e = (e_1, \dots, e_n)$ be a basis of U. The matrix of $T : U \to U$ in basis e is*

$$[T]_e = \big([T(e_1)]_e, \dots, [T(e_n)]_e\big).$$

The columns are the coordinate vectors of the images of the basis vectors.

Example. *For $T = T_A$ on $\mathbb{R}^2$ with the standard basis, $[T]_e = A$.*

Example (Derivative matrix for P_2)**.** *Working in P_2 with basis $e = (1, x, x^2)$, every polynomial $a + bx + cx^2$ has coordinate vector $(a, b, c)^T$. To build $[D]_e$, apply D to each basis vector and write the result's coordinates as a column:*

$$D(1) = 0 \to \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad D(x) = 1 \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad D(x^2) = 2x \to \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix},$$

so $[D]_e = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$. *To verify:* $f = 3 + 5x - 2x^2$ *has* $f' = 5 - 4x$, *and indeed*

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 0 \end{pmatrix} \longleftrightarrow 5 - 4x. \checkmark$$

Example (Matrix of T in a non-standard basis)**.** *Let* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *be defined by* $T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix}$, *and let* $f = (f_1, f_2)$ *with* $f_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. *Find* $[T]_f$.

Column 1. $T(f_1) = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4f_1 + 0f_2$, *so* $[T(f_1)]_f = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$.

Column 2. $T(f_2) = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 0f_1 + 2f_2$, *so* $[T(f_2)]_f = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

Therefore $[T]_f = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$, *which is diagonal.* f_1 *and* f_2 *are eigenvectors of* T *with eigenvalues 4 and 2.*

6 Isomorphisms and Coordinates

February 23, 2026

Definition 45 (Isomorphism)**.** $T : U \to V$ *is an isomorphism if it is linear and bijective.*

Remark 14 (How to check T is an isomorphism)**.** *In practice, verify the following three things:*

1. ***Linearity****: check* $T(u + u') = T(u) + T(u')$ *and* $T(\lambda u) = \lambda T(u)$.
2. ***Injectivity****: show* $\ker(T) = \{0\}$, *i.e., the only solution to* $T(u) = 0$ *is* $u = 0$. *This is equivalent to saying* T *does not collapse any nonzero vector to zero.*
3. ***Surjectivity****: show* $\mathrm{Im}(T) = V$, *i.e., every* $v \in V$ *can be written as* $T(u)$ *for some* $u \in U$.

A useful shortcut: if $\dim U = \dim V = n$, *you only need to check linearity and* one *of injectivity or surjectivity — the other follows automatically by rank-nullity.*

Theorem 10 (Properties of Isomorphisms)**.** *Let* $T : V \to V$ *be a linear map and let* $A = [T]_e$ *be the matrix of* T *in some basis* e *of* V. *Then the following are equivalent:*

1. T *is an isomorphism.*
2. A *is invertible.*

3. $\det(A) \neq 0$.

Theorem 11 (Isomorphism Theorem)**.** *$U \cong V \iff \dim U = \dim V$. In particular, every n-dimensional space is isomorphic to k^n.*

Example. *$P_8 \cong \mathbb{R}^9 \cong M_3(\mathbb{R})$ since all have dimension 9.*

6.1 The Canonical Isomorphism

Definition 46 (Coordinate map)**.** *Given vectors $v_1, \ldots, v_n \in V$, define $\Phi : k^n \to V$ by*

$$\Phi \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \lambda_1 v_1 + \cdots + \lambda_n v_n.$$

Then $\operatorname{Im}(\Phi) = \operatorname{span}(v_1, \ldots, v_n)$.

Theorem 12 (Isomorphism Theorem 2)**.** *Φ is an isomorphism $\iff (v_1, \ldots, v_n)$ is a basis for V. In this case, $[v]_e = \Phi^{-1}(v)$.*

6.2 Change of Basis

February 27 – March 13, 2026

6.2.1 Change of Basis Matrix

Let $e = (e_1, \ldots, e_n)$ and $f = (f_1, \ldots, f_n)$ be two bases of V, and let M_e and M_f denote the matrices whose columns are the basis vectors of e and f written in standard coordinates. The change of basis matrices are:

- $Q = M_e^{-1} M_f$: converts f-coordinates to e-coordinates.
- $P = M_f^{-1} M_e = Q^{-1}$: converts e-coordinates to f-coordinates.

Remark 15. *When e is the standard basis, $M_e = I$, so the formulas simplify to $Q = M_f$ and $P = M_f^{-1}$.*

Example (Building Q when neither basis is standard)**.** *Let $e = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$ and $f = \left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$. Find Q.*

$$M_e = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_f = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad M_e^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$Q = M_e^{-1} M_f = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad P = Q^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

6.2.2 Change of Basis Formula

Theorem 13 (Change of basis)**.** *If $A = [T]_e$ and $B = [T]_f$, then*

$$B = PAQ, \quad P = Q^{-1}.$$

Remark 16 (The $Q^{-1}AQ$ shortcut)**.** *Since $P = Q^{-1}$, the formula is exactly $B = Q^{-1}AQ$. In practice: build Q (columns = new basis vectors, or $M_e^{-1}M_f$ in general), then compute $Q^{-1}AQ$.*

Example (Full change of basis computation)**.** *Let e be the standard basis of $\mathbb{R}^2$, $f_1 = \begin{pmatrix}1\\3\end{pmatrix}$, $f_2 = \begin{pmatrix}2\\5\end{pmatrix}$, and $T : \begin{pmatrix}x\\y\end{pmatrix} \mapsto \begin{pmatrix}2x-7y\\4x-3y\end{pmatrix}$. Find $[T]_f$.*

Step 1: $Q = \begin{pmatrix}1 & 2\\3 & 5\end{pmatrix}$.

Step 2: $P = Q^{-1} = \begin{pmatrix}-5 & 2\\3 & -1\end{pmatrix}$.

Step 3:

$$AQ = \begin{pmatrix}2 & -7\\4 & -3\end{pmatrix}\begin{pmatrix}1 & 2\\3 & 5\end{pmatrix} = \begin{pmatrix}-19 & -31\\-5 & -7\end{pmatrix}, \quad B = PAQ = \begin{pmatrix}85 & 141\\-52 & -86\end{pmatrix}.$$

7 Diagonalization

March 13–16, 2026

7.1 Eigenvalues and Eigenvectors

Definition 47 (Eigenvalue and eigenvector)**.** *$\lambda \in k$ is an eigenvalue of T if $\exists\, v \neq 0$ with $Tv = \lambda v$; v is an eigenvector. The eigenspace is*

$$E_\lambda = \{v \in V : Tv = \lambda v\}.$$

Proposition 15. *E_λ is a subspace of V.*

Proposition 16. *If $\lambda_1 \neq \lambda_2$, then $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.*

Proof. If $v \in E_{\lambda_1} \cap E_{\lambda_2}$, then $\lambda_1 v = Tv = \lambda_2 v$, so $(\lambda_1 - \lambda_2)v = 0$. Since $\lambda_1 \neq \lambda_2$, we get $v = 0$. □

7.2 The Characteristic Polynomial

Definition 48 (Characteristic polynomial)**.**

$$p(\lambda) = \det(A - \lambda\,\mathrm{Id}).$$

Theorem 14 (Characterization of Eigenvalues)**.** *λ is an eigenvalue of $A \iff \lambda$ is a root of $p(\lambda)$.*

March 18, 2026

7.3 Algebraic and Geometric Multiplicity

Definition 49 (Algebraic and geometric multiplicity)**.** *For an eigenvalue λ of A:*

- *The algebraic multiplicity (AM) of λ is its exponent as a root of $p(\lambda)$.*
- *The geometric multiplicity (GM) of λ is $\dim E_\lambda = \dim \ker(A - \lambda \operatorname{Id})$.*

Theorem 15 (AM-GM inequality)**.** *For every eigenvalue λ:*

$$1 \leq \mathrm{GM}(\lambda) \leq \mathrm{AM}(\lambda).$$

A is diagonalizable $\iff$ $\mathrm{GM}(\lambda) = \mathrm{AM}(\lambda)$ for every eigenvalue, i.e.,

$$V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}.$$

Remark 17. *In particular, if A has n distinct eigenvalues (all $AM = 1$), then A is automatically diagonalizable.*

7.4 Diagonalization

Definition 50 (Diagonalizable)**.** *T (or its matrix A) is diagonalizable if $\exists$ a basis $e = (e_1, \ldots, e_n)$ and scalars $\lambda_1, \ldots, \lambda_n$ with $T(e_i) = \lambda_i e_i$.*

Method 2 (How to diagonalize A.)**.**

1. *Compute $p(\lambda) = \det(A - \lambda \operatorname{Id})$ and find its roots (the eigenvalues).*
2. *For each eigenvalue λ_i, find $\ker(A - \lambda_i \operatorname{Id})$ to get eigenvectors.*
3. *If there are n linearly independent eigenvectors, write them as columns of Q; then $Q^{-1}AQ$ is diagonal.*

Example (Diagonalization of a 4×4 matrix)**.** *Diagonalize* $A = \begin{pmatrix} 5 & -2 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix}$.

Characteristic polynomial. *A is block diagonal:*

$$p(\lambda) = (5-\lambda)(3-\lambda) \cdot \big[(1-\lambda)(4-\lambda) - 4\big] = -\lambda(\lambda-3)(\lambda-5)^2.$$

Eigenvalues: $\lambda = 0$ ($AM = 1$), $\lambda = 3$ ($AM = 1$), $\lambda = 5$ ($AM = 2$).

Eigenvectors: *For $\lambda = 5$: $E_5 = \operatorname{span}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}$ ($GM = 2 = AM$ ✓). For $\lambda = 3$: $E_3 = \operatorname{span}\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$. For $\lambda = 0$: $E_0 = \operatorname{span}\begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \end{pmatrix}$.*

$$P = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 2 & 0 & 1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Example (3×3 diagonalization)**.** *Let* $A = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 4 & 3 \\ 2 & 0 & 1 \end{pmatrix}$. *Expanding along the second column:*

$$p(\lambda) = (4-\lambda)\det\begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} = (\lambda-4)(\lambda-3)(\lambda+1).$$

Three distinct eigenvalues $\Rightarrow$ *diagonalizable. Eigenvectors:* $f_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ -6 \\ 1 \end{pmatrix}$, $f_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

$$Q = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -6 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad A = QDQ^{-1}.$$

7.5 Similar Matrices

Definition 51 (Similar matrices)**.** *A and B are* ***similar*** *if there exists an invertible Q with $A = Q^{-1}BQ$.*

Problem 22 (Similar matrices have equal characteristic polynomials)**.** *Let A be similar to B, i.e., $A = Q^{-1}BQ$ for some invertible Q. Prove $p_A = p_B$.*

Problem 23 (Characteristic polynomial and invertibility)**.** *Let $A \in M_n(\mathbb{R})$ be non-invertible. Show $p(\lambda)$ has a root at $\lambda = 0$.*

8 Inner Product Spaces

March 18, 2026

An inner product space is a vector space V equipped with an inner product — a way to measure length, angle, and perpendicularity.
Let V be a vector space over $K = R$ or $K = C$. We aim to define:

1. the norm of a vector $v \in V$, which is a real number $||v||$ representing the length of v;
2. what it means for two vector $u, v \in V$ to be perpendicular (or orthogonal), denoted $u \perp v$.

One can do both by means of an inner product on V. We explain the case $K = R$ first.

8.1 Real Inner Products

Definition 52 (Euclidean Norm)**.** *The Euclidean norm of a vector* $v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ *is*

$$\|v\| = \sqrt{x_1^2 + \cdots + x_n^2} = \sqrt{\langle v, v \rangle},$$

where $\langle v, v \rangle$ *is the standard dot product of* v *with itself.*

Example (Frobenius Norm on $M_n(\mathbb{R})$)**.** *The matrix analogue of the Euclidean norm is the Frobenius norm. For* $A = (a_{ij})$ *and* $B = (b_{ij})$ *in* $M_n(\mathbb{R})$*, define the inner product* $\langle A, B \rangle = \sum_{i,j} a_{ij} b_{ij}$*, which induces the norm*

$$\|A\| = \sqrt{\langle A, A \rangle} = \sqrt{\sum_{i,j} a_{ij}^2}.$$

This is the direct analogue of the Euclidean norm, summing the squares of all entries instead of all components.

Definition 53 (Inner product)**.** *Let* V *be a real vector space. An inner product on* V *is a map* $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ *satisfying:*

1. ***Linearity on the left****:* $\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle$ *and* $\langle \lambda v, w \rangle = \lambda \langle v, w \rangle$*.*
2. ***Symmetry****:* $\langle u, v \rangle = \langle v, u \rangle$*.*
3. ***Positive definiteness****:* $\langle v, v \rangle \geq 0$*, and* $\langle v, v \rangle = 0 \Rightarrow v = 0$*.*

A vector space with an inner product is called an ***inner product space****.*

Remark 18. *By symmetry, linearity on the left implies linearity on the right:* $\langle u, v_1 + v_2 \rangle = \langle u, v_1 \rangle + \langle u, v_2 \rangle$ *and* $\langle u, \lambda v \rangle = \lambda \langle u, v \rangle$*.*

Definition 54 (Norm and Perpendicularity)**.** *Let* $\langle \cdot, \cdot \rangle$ *be an inner product on* V*.*

1. *The norm of* $v \in V$ *is defined by*

$$||v|| := \sqrt{\langle v, v \rangle}.$$

2. *We say that* u *and* v *are perpendicular or orthogonal and write* $u \perp v$ *if*

$$\langle u, v \rangle = 0.$$

Remark 19 (Length vs Distances)**.** *The norm of* v *represents the length of the vector* v*. To compute the distance* d *between two points* u *and* v *in* V *(extremities of vectors), one needs to compute:*

$$dist(u, v) = ||u - v||$$

Example (Standard inner product on $\mathbb{R}^n$)**.** *For $v = (x_1, \ldots, x_n)^T$ and $w = (y_1, \ldots, y_n)^T$, the dot product $\langle v, w\rangle = x_1y_1 + \cdots + x_ny_n$ is an inner product.*

Example (Function space inner product)**.** *On $V = C^0([0,1])$, define $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$. This satisfies all three axioms; positive definiteness uses the fact that if $f^2 \geq 0$ is continuous and $\int f^2 = 0$, then $f = 0$ everywhere.*

Proposition 17 (Norm expansion)**.** *For any $u, v \in V$:*

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2.$$

Proof. $\langle u+v, u+v\rangle = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$. □

Theorem 16 (Pythagoras)**.** $u \perp v \iff \langle u, v\rangle = 0 \iff \|u \pm v\|^2 = \|u\|^2 + \|v\|^2$.

8.2 Complex inner products

Definition 55 (Inner product on $\mathbb{C}^n$)**.** *For $x, y \in \mathbb{C}^n$, define*

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i\overline{y_i}.$$

Then $\|x\|^2 = \sum_{i=1}^n |x_i|^2 \geq 0$, with equality iff $x = 0$.

Definition 56. *An inner product on a complex vector space V is a function assigning a complex number $\langle u, v\rangle$ to $u, v \in V$: $V \times V \to \mathbb{C}$, $(u, v) \to \langle u, v\rangle$ satisfying:*

1. ***Linearity on the left**: $\langle v_1 + v_2, w\rangle = \langle v_1, w\rangle + \langle v_2, w\rangle$ and $\langle \lambda v, w\rangle = \lambda\langle v, w\rangle$, $\forall u, v, w \in V, \lambda \in \mathbb{C}$.*
2. ***Symmetry**: $\langle u, v\rangle = \overline{\langle v, u\rangle}$.*
3. ***Positive definiteness**: $\langle v, v\rangle \geq 0$, and $\langle v, v\rangle = 0 \Rightarrow v = 0$.*

Proposition 18 (Complex symmetry)**.** *Complex inner products are skew-linear in the second variable. This is also called anti-linear and means*

- $\langle w, u + v\rangle = \langle w, u\rangle + \langle w, v\rangle$
- $\langle w, \lambda u\rangle = \overline{\lambda}\langle w, u\rangle$

Definition 57 (Adjoint Matrix)**.** *Let $A \in M_n(\mathbb{C})$ be a complex matrix. The adjoint of A is the complex matrix $A^* \in M_n(\mathbb{C})$ defined by $A^* = \overline{A}^t$.*

8.3 Cauchy-Schwarz Inequality

Theorem 17 (Cauchy-Schwarz Inequality)**.** *For any u, v in an inner product space V:*

$$|\langle u, v\rangle| \leq \|u\| \, \|v\|,$$

with equality if and only if u and v are linearly dependent.

Proof. If $v = 0$ both sides are 0. Otherwise, let $w = u - \frac{\langle u,v\rangle}{\|v\|^2} v$ (the component of u orthogonal to v). Then $\langle w, v\rangle = 0$, and by positive definiteness:

$$0 \leq \|w\|^2 = \left\| u - \frac{\langle u, v\rangle}{\|v\|^2} v \right\|^2 = \|u\|^2 - \frac{|\langle u, v\rangle|^2}{\|v\|^2},$$

which rearranges to $|\langle u, v\rangle|^2 \leq \|u\|^2 \|v\|^2$. □

Theorem 18 (Triangle inequality)**.** $\|u + v\| \leq \|u\| + \|v\|$.

Proof. $\|u + v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$. □

8.4 Orthonormal Bases

March 23, 2026

Definition 58 (Orthogonal)**.** *A family $v_1, \ldots, v_n$ of vectors is orthogonal if*

$$\langle v_i, v_j\rangle = 0, if\ i \neq j.$$

Definition 59 (Orthonormal)**.** *A family $v_1, \ldots, v_n$ of vectors is orthonormal if it is orthogonal and $||v_i|| = 1$, $\forall i = 1, \ldots, n$. Basically,*

$$\langle v_i, v_j\rangle = \delta_{ij}$$

Definition 60 (Orthonormal Basis)**.** *An orthonormal basis (ONB) is a basis which is orthonormal.*

Proposition 19. *Any orthogonal set of nonzero vectors is linearly independent.*

Proof. Suppose $\sum \lambda_i e_i = 0$. Taking the inner product with e_k:

$$0 = \left\langle e_k, \sum_i \lambda_i e_i \right\rangle = \lambda_k \|e_k\|^2,$$

so $\lambda_k = 0$ since $\|e_k\| \neq 0$. □

Proposition 20 (Coordinates in an ONB)**.** *If $(e_1, \ldots, e_n)$ is an ONB of V, then for any $v \in V$:*

$$v = \sum_{i=1}^{n} \langle v, e_i\rangle \, e_i.$$

In other words, the i-th coordinate of v in the ONB is simply $\langle v, e_i\rangle$.

Proof. Write $v = \sum \lambda_i e_i$. Taking $\langle \cdot, e_k \rangle$ of both sides: $\langle v, e_k \rangle = \lambda_k \|e_k\|^2 = \lambda_k$. $\square$

Definition 61 (Orthogonal Matrix)**.** *A matrix $A \in M_n(\mathbb{R})$ is orthogonal if its columns form an ONB of $\mathbb{R}^n$ (endowed with the standard dot product).*

Proposition 21. *Let $TX = AX$ be the linear transformation of $\mathbb{R}^n$ associated with matrix $A \in M_n(\mathbb{R})$. Then A is an orthogonal matrix iff T preserves the dot product in the sense that:*

$$\langle TX, TY \rangle = \langle X, Y \rangle,$$

for all vectors $X, Y \in \mathbb{R}^n$, where $\langle X, Y \rangle := X \cdot Y$ is the dot product.

Definition 62 (Orthogonal Matrix 2)**.** *A matrix A is said to be orthogonal if $AA^t = A^t A = Id$; equivalently, A is invertible with inverse $A^{-1} = A^t$.*

Definition 63 (Unitary Matrix)**.** *A matrix $A \in M_n(\mathbb{C})$ is unitary if its columns form an ONB of $\mathbb{C}$.*

Definition 64 (Unitary Matrix 2)**.** *A complex $A \in M_n(\mathbb{C})$ is unitary if $AA^* = A^* A = Id$; equivalently, A is invertible with inverse $A^{-1} = A^*$.*

8.5 Symmetric Matrices Are Diagonalizable

March 25, 2026

Theorem 19 (Spectral theorem - version 1)**.** *Symmetric matrices are diagonalizable.*

Example. *Let the matrix* $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. *Then A is symmetric and thus diagonalizable.*

Theorem 20 (Spectral theorem - version 2)**.** *If $A \in M_n(\mathbb{R})$ is symmetric, then there exists an ONB of eigenvectors.*

Lemma 21. *If A is symmetric, and $\lambda_1 \neq \lambda_2$ are two different eigenvalues of A, then $E_{\lambda_1} \perp E_{\lambda_2}$.*

Theorem 22 (Spectral theorem - version 3)**.** *If $A \in M_n(\mathbb{R})$ is symmetric, there exists an orthogonal matrix Q and a diagonal matrix D such that*

$$A = QDQ^t.$$

8.6 Orthogonal Matrices in Dimension 2

Every 2×2 orthogonal matrix is one of two types: a rotation or a reflection.

Proposition 22. *Let $A \in M_2(\mathbb{R})$ be orthogonal. Then A is either a rotation*

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

about the origin by angle θ, or a reflection

$$S_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

across a line through the origin at angle $\theta/2$. In symbols:

$$O(2) = \{R_\theta : \theta \in \mathbb{R}\} \cup \{S_\theta : \theta \in \mathbb{R}\}.$$

Remark 20. *You can tell them apart by the determinant:*

$$\det(R_\theta) = \cos^2\theta + \sin^2\theta = 1, \qquad \det(S_\theta) = -\cos^2\theta - \sin^2\theta = -1.$$

So rotations have determinant 1 and reflections have determinant -1.

Remark 21 (Exam tip)**.** *The inverse of any orthogonal matrix is just its transpose: $A^{-1} = A^T$. No row reduction needed. In particular, $R_\theta^{-1} = R_{-\theta}$ (rotate back) and $S_\theta^{-1} = S_\theta$ (reflecting twice gives the identity).*

8.7 Isomorphism Theorem For Inner Product Spaces

Theorem 23 (Isomorphism Theorem For Inner Product Spaces)**.** *Every finite-dimensional inner product space admits an ONB.*

Theorem 24. *Every inner product on a finite-dimensional space over K is isomorphic to the standard dot product on K^n.*

8.8 Gram-Schmidt Orthonormalization

Given two linearly independent vectors v_1, v_2, Gram-Schmidt produces an ONB $\{e_1, e_2\}$ with $\operatorname{span}(e_1, e_2) = \operatorname{span}(v_1, v_2)$ in two steps:

1. Normalize v_1: set $e_1 = \dfrac{v_1}{\|v_1\|}$.
2. Subtract from v_2 its projection onto e_1, then normalize:

$$w_2 = v_2 - \langle v_2, e_1\rangle\, e_1, \qquad e_2 = \frac{w_2}{\|w_2\|}.$$

The key idea is that $\langle v_2, e_1\rangle$ is the projection coefficient, telling you how much of v_1's direction is in v_2. Subtracting it off leaves a vector orthogonal to e_1.

Example. *Let* $v_1 = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$ *and* $v_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.
Step 1: $e_1 = \dfrac{v_1}{\|v_1\|} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.
Step 2: $\langle v_2, e_1 \rangle = 3 \cdot 1 + 2 \cdot 0 = 3$, *so*

$$w_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \qquad e_2 = \frac{w_2}{\|w_2\|} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

So $\{e_1, e_2\} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ *is an ONB for* $\mathbb{R}^2$.

9 Appendix

9.1 Proof: Standard Basis is a Basis

Claim 1. $e_1, \ldots, e_n$ span $\mathbb{R}^n$.

Proof. $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 e_1 + \cdots + x_n e_n.$ □

Claim 2. $e_1, \ldots, e_n$ are linearly independent.

Proof. $\lambda_1 e_1 + \cdots + \lambda_n e_n = 0 \implies \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = 0 \implies \lambda_i = 0.$ □

10 Solutions

Solution 1 (Hermitian Matrix)**.** $(AB)^* = (\overline{AB})^T = (\overline{A}\,\overline{B})^T = \overline{B}^T \overline{A}^T = B^* A^*$.

Solution 2 (Hermiticity and Anti-Hermiticity)**.** *(⇒): Suppose* A *is Hermitian, then* $A = \overline{A}^t$, *then* $(iA)^* = \overline{iA^t} = -iA^* = -iA$. *Therefore,* $-(iA)^* = iA$.
(⇐): Suppose $iA = -(iA)^*$, *then* $iA = -(iA)^* = -(-iA^*) = iA^* = iA$. *Therefore,* $iA^* = iA \implies A = A^*$.

Solution 3 (Invertibility)**.** *Suppose* B *and* B' *are both inverses of* A. *Then* $B = BI = B(AB') = (BA)B' = IB' = B'$.

Solution 4 (Invertibility 2)**.** *Suppose the matrix is invertible. Then* $\det \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0 \cdot 0 - 1 \cdot 0 = 0$, *but an invertible matrix must have nonzero determinant. Contradiction.*

Solution 5 (Span)**.** $\operatorname{span}(v_1, v_2) = \left\{ \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix} : x, y \in \mathbb{R} \right\}$.

Solution 6 (Span 2)**.** *Setting up* $u = au_1 + bu_2 + cu_3$ *gives the augmented matrix* $\left(\begin{array}{ccc|c} 3 & 1 & 2 & 2 \\ 10 & 3 & 8 & 10 \\ 7 & -2 & 1 & 7 \end{array}\right)$. *Gaussian elimination gives* $a = \frac{2}{21}$, $b = -\frac{46}{21}$, $c = \frac{41}{21}$, *so* $u \in \operatorname{span}(u_1, u_2, u_3)$.

Solution 7 (Subspace criterion)**.** $U = \{x \in k^n : Ax = 0\}$. *Check:* $A0 = 0$ *so* $0 \in U$*; if* $Ax = Ay = 0$ *then* $A(x + y) = 0$*; if* $Ax = 0$ *then* $A(\lambda x) = 0$. *So* U *is a subspace.*

Solution 8 (Subspace criterion 2)**.** $U = \{Ax : x \in k^n\}$. *Check:* $A0 = 0$*;* $Ax_1 + Ax_2 = A(x_1 + x_2)$*;* $\lambda Ax = A(\lambda x)$. *So* U *is a subspace.*

Solution 9 (Subspace criterion 3)**.** *If* $V \neq \{0\}$*, pick* $v \in V$, $v \neq 0$. *For any* $x \in k$*, write* $x = \lambda v$ *for* $\lambda = x/v \in k$. *By closure under scalar multiplication,* $x \in V$. *Hence* $V = k$.

Solution 10 (Subspace criterion 4)**.** $0 = 0v_1 + \cdots + 0v_n \in \operatorname{span}$*; sums and scalar multiples of linear combinations are linear combinations. Hence* $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace.*

Solution 11 (Span membership)**.** $x = 3$, $y = 1$, $z = -2$, *so* $v = 3v_1 + v_2 - 2v_3 \in \operatorname{span}(v_1, v_2, v_3)$.

Solution 12 (Span membership 2)**.** *Same system:* $t = 3$, $y = 1$, $z = -2$, *so* $f = 3f_1 + f_2 - 2f_3$.

Solution 13 (Infinite dimension of P_∞)**.** *If* $\dim P_\infty < \infty$*, some* $f_1, \ldots, f_n$ *span* P_∞. *Let* $d = \max \deg(f_i)$. *Then every polynomial in the span has degree* $\leq d$*, but* x^{d+1} *does not. Contradiction.*

Solution 14 (Linear independence)**.** $3v_1 + 5v_2 - 2v_3 = 0$*, so the vectors are not LI.*

Solution 15 (Coordinate vectors)**.** $\lambda_1 = 2$, $\lambda_3 = 0$, $\lambda_2 = -\frac{1}{3}$. *So* $[g]_S = \begin{pmatrix} 2 \\ -1/3 \\ 0 \end{pmatrix}$.

Solution 16 (Coordinate vectors 2)**.** *Evaluate:* $f(1) = -1$, $f(2) = 5$, $f(3) = 15$. *So* $[f]_e = \begin{pmatrix} -1 \\ 5 \\ 15 \end{pmatrix}$.

Solution 17 (Finding a Basis)**.** *Form the matrix with* v_1, v_2, v_3, v_4 *as columns and row reduce:*

$$\begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 8 \\ 1 & 3 & 0 & 2 \\ 1 & 2 & 1 & 10 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & 4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -3 & 4 \\ 0 & 0 & 0 & 8 \end{pmatrix}.$$

Pivots are in columns 1, 2, 3, 4, so all four vectors are LI and $\{v_1, v_2, v_3, v_4\}$ is a basis of V.

Solution 18 (Finding a Basis 2)**.** *Use the isomorphism* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ *to get coordinate vectors* $\begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 5 \\ 7 \end{pmatrix}$. *Row reducing the matrix with these as columns gives a free variable for M_3, revealing $M_3 = 2M_1 + M_2$. So $\{M_1, M_2\}$ is a basis.*

Solution 19 (Finding a Basis 3)**.** *(1) $A = A^*$ means $a_{ij} = \overline{a_{ji}}$. The diagonal entries satisfy $a_{ii} = \overline{a_{ii}}$, so they are real. The off-diagonal entries satisfy $a_{12} = \overline{a_{21}}$. Writing $z = a_{12}$ gives the form* $\begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}$.

(2) Writing $z = a + bi$ with $a, b \in \mathbb{R}$, the general Hermitian matrix is

$$x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + a \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}.$$

These four matrices are LI and span U, so they form a basis. Hence $\dim_{\mathbb{R}} U = 4$.

Solution 20 (Finding a Basis 4)**.** *Since $f \in U$ is even, it is determined by its values at $x = 0, 1, 2$, so $\dim U = 3$. It suffices to show S is LI (then $|S| = \dim U = 3$ implies it is a basis). Suppose $c_1 g_1 + c_2 g_2 + c_3 g_3 = 0$. Evaluating:*

$$x = 0: \ c_1 = 0, \qquad x = 1: \ c_1 + 0 + c_3 = 0 \implies c_3 = 0, \qquad x = 2: \ c_1 + 0 + 0 = 0.$$

(One checks $g_2(1) = 0$, $g_2(2) = 0$, $g_3(1) = 1$, $g_3(2) = 0$.) Hence $c_1 = c_2 = c_3 = 0$ and S is LI, so S is a basis.

Solution 21 (Basis Extension)**.** *If L spans V, done. Otherwise, since L is LI but not spanning, there exists $v_1 \in V \setminus \operatorname{span}(L)$. Then $L \cup \{v_1\}$ is LI. Repeating: at each step we add one vector to L while maintaining LI. Since V is finite-dimensional, this process terminates in at most $\dim V - |L|$ steps, yielding a basis $B \supseteq L$.*

Solution 22 (Similar matrices have equal characteristic polynomials)**.** $p_A(\lambda) = \det(A - \lambda I) = \det(Q^{-1}BQ - \lambda I) = \det(Q^{-1}(B - \lambda I)Q) = \det(Q^{-1}) \det(B - \lambda I) \det(Q) = p_B(\lambda)$.

Solution 23 (Characteristic polynomial and invertibility)**.** $\det(A) = 0$ *since A is non-invertible, so $p(0) = \det(A - 0 \cdot I) = \det(A) = 0$. Hence 0 is a root of $p(\lambda)$.*

11 Useful Links

Proofs: https://www.youtube.com/watch?v=V5tUc-J124s